

|               | Wi-Fi 4        | Wi-Fi 5                      | Wi-Fi 6                      | Wi-Fi 6E | Wi-Fi 7              |
|---------------|----------------|------------------------------|------------------------------|----------|----------------------|
| Launch date   | 2007           | 2013                         | 2019                         | 2021     | 2024 (expected)      |
| IEEE standard | 802.11n        | 802.11ac                     | 802.11ax                     |          | 802.11be             |
| Max data rate | 1.2 Gbps       | 3.5 Gbps                     | 9.6 Gbps                     |          | 46 Gbps              |
| Bands         | 2.4 GHz, 5 GHz | 5 GHz                        | 2.4 GHz, 5 GHz               | 6 GHz    | 2.4 GHz, 5 GHz, 6 Hz |
| Channel size  | 20, 40 MHz     | 20, 40, 80<br>80+80, 160 MHz | 20, 40, 80<br>80+80, 160 MHz |          | Up to 320 MHz        |
| Modulation    | 64-QAM         | 256-QAM                      | 1024-QAM                     |          | 4096-QAM             |
| MIMO          | 4x4 MIMO       | 4x4 MIMO, DL MU-MIMO         | 8x8 UL/DL MU-MIMO            |          | 16x16 UL/DL MU-MIMO  |

is it assumed to be a revolution in the Wi-Fi stream.

For starters Wi-Fi 7 is 4.8X faster than Wi-Fi 6 and 13X faster than Wi-Fi 5 and it executes extremely high throughput of 46 Gbps. Compared to that of the other devices Wi-Fi 7 works on all three frequency bands, 2GHz, 5GHz and 6GHz.

Wi-Fi 6 comes in bearing solutions to reduce latency, increase the overall capacity and boost the stability and the efficiency of the device communication to prevent any buffering, lag or congestion.

The fundamental features that are highlighted for the device are the introduction of 320Hz ultra-wide bandwidth. This is jump from the previous 160Hz bandwidth of Wi-Fi 6. In addition the device has to offer a whopping 4096 QAM (Quadrature Amplitude Modulation). For everyone who is not familiar with this concept, QAM can be understood as the medium of transferring data from point A to B. Now the advancement is the rate of speed the transfer takes place at and the data that can be transferred at one. Wi-Fi allows 120 per cent more data transfer with higher speed, suggested 46Gbps.

Wi-Fi 7 also offers Multi-Link Operation, as the name suggest, the device is capable of aggregating more than two frequencies simultaneously for sending and receiving data. This feature works across frequencies wherein Wi-Fi holds a total of three frequencies to operate on.

The device will also sport Multi - RU for a single user, wherein various resource units will come to together on a stream and the user can switch and use multiple units on the device

without any hindrance. Moreover, the device has to offer preamble puncturing where every available spectrum can be used on multiple width portions of bandwidth can be transmitted without any interference. Lastly, the device has to offer 16x16 MU-MIMO. In simpler terms there are 16 streams that the device has to offer wherein the device can send out streams to multiple devices at the same time.

## Adoption In India

Having explained all the different aspects and workings of Wi-Fi 7, it is safe to say that it will be a definite

with devices that support the high-intensity network stream and we are still a bit far away from that.

However, that does not in any way mean that India is behind on attaining the technology. There are still approvals to be drawn by the Federal Communications Commission and the launch for Wi-Fi 7 can be expected by the end of 2024. There have been rumours of laptops, smartphones and PCs being developed to collaborate on the technology.

Wi-Fi 7 is some time away from launch. There has been news of Intel and Qualcomm coming together for



quality of life improvement for many. However, don't run out to your local electronics store just yet.

Wi-Fi 7 was introduced earlier this year in US and since there have been speculations of when the device will be available in India and whether the systems in India are compatible for the device. Wi-Fi 7 requires to be connected

preliminary tests of the technology, and are also reportedly working on developing a cross-vendor compatible solution. Moreover, the first wave of Wi-Fi 7 devices are expected to hit the shelves in India by 2024. So, there is time. But, things do look promising. Let us know what you think about Wi-Fi 7 by writing to us at [editor@digit.in](mailto:editor@digit.in). 